CONTINUOUS-STATE CONSENSUS

Continuous-State Consensus Redefines Programmable Value

The Continuous-Wave State Layer

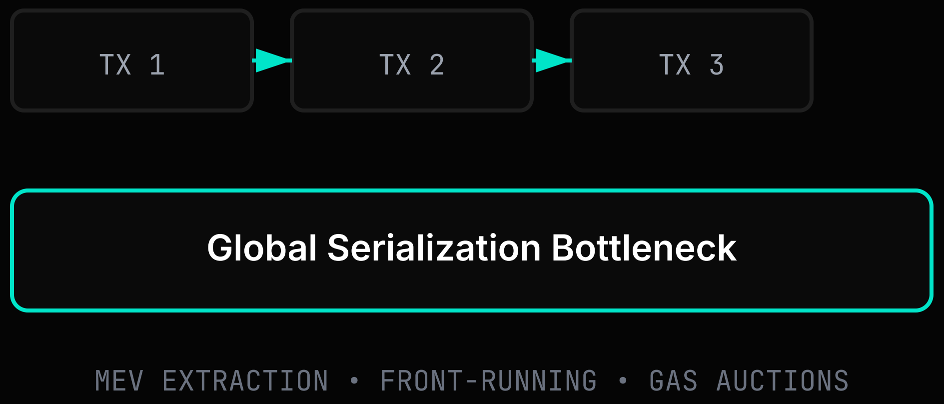
Financial state as a continuous wave-field. Commutative operations superpose in parallel. Stateful operations retain strict causal ordering. Mathematical certainty replaces sequential bottlenecks.

Rust Reference Implementation — NTT Consensus — Vector-Clock DAG

THE PROBLEM

Linear Transaction Ordering Becomes the Latency Floor

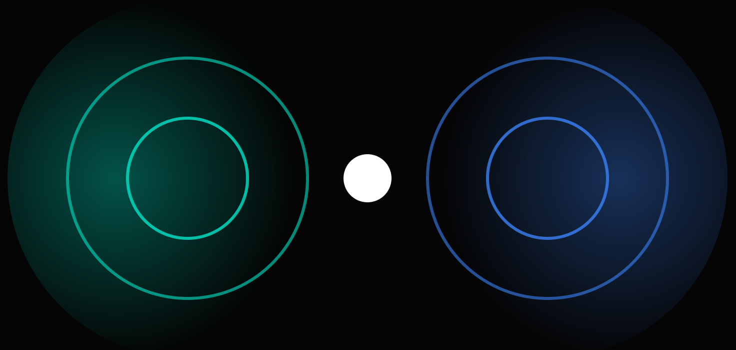
- **Sequential execution** forces every operation to wait for global serialization.
- **MEV and front-running** are direct consequences of artificial time-ordering.
- **Throughput scales only** by making blocks bigger or chains faster—both hit hard physical limits.



THE SOLUTION

Wave-Field Synthesis Enables Infinite Concurrency

- **Accounts are coordinate frequencies** in a shared state field.
- **State transitions are signed phase-shifts** that superpose naturally, like interfering waves.
- **Commutative inputs blend simultaneously** without requiring a canonical arrival order.

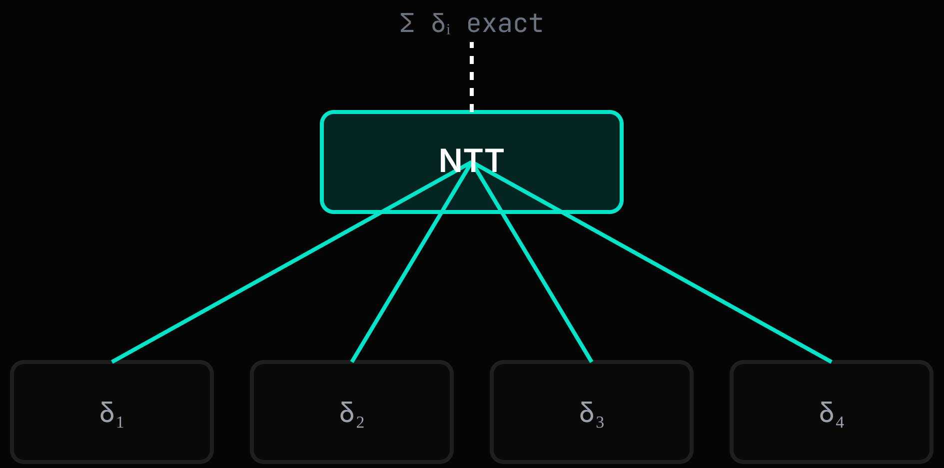


CONSTRUCTIVE INTERFERENCE = SYNTHESIZED STATE

THE TECHNOLOGY

Number-Theoretic Transforms Guarantee Exact Algebraic Consensus

- **Commutative operations** are aggregated via NTT over prime field `P = 998,244,353`.
- **Forward and inverse Cooley-Tukey butterflies** recover the exact algebraic total with no floating-point drift.
- **Stateful operations** remain strictly ordered through a vector-clock DAG before entering the wave layer.



NO ROUNDING • NO DRIFT • INTEGER MODULAR ARITHMETIC

TECHNICAL VALIDATION

100,000 Concurrent Updates Synthesize to a Single Exact Result

- **Stress test:** 100,000 threads update the same localized 2048-bin coordinate simultaneously.
- **NTT/INTT round-trip** recovers the algebraic total exactly: 500,000 = 500,000.
- **Mesh resilience:** six-node local network survives disconnection of two nodes (~33%) and resumes consensus after reconnection.

100k
Concurrent operations targeting the same spectrum bin

0
Rounding error after forward/inverse NTT

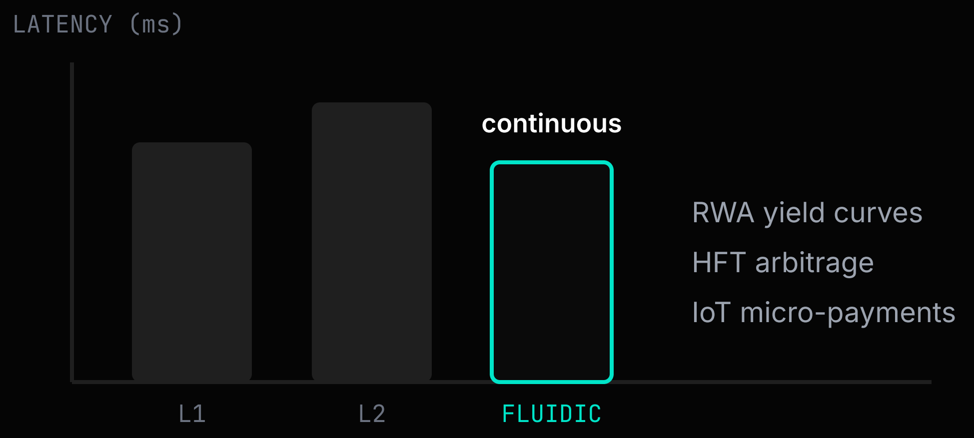
4/6
Nodes maintain consensus during partition

<1%
Metabolic decay engine overhead under load

THE MARKET

High-Frequency RWA and Agent Swarms Require Continuous Settlement

- **Tokenized treasuries, FX, and derivatives** need microsecond-level price updates, not block-time batching.
- **Autonomous agent swarms** generate thousands of commutative micro-transactions per second.
- **Existing L1/L2 architectures** force these workloads into artificial queues, increasing cost and latency.



BUSINESS MODEL

Spectrum Leasing and Metabolic Burn Align Economic Incentives

SPECTRUM LEASING

Applications stake WAVE tokens to lease contiguous, low-noise frequency bands in the state-field spectrum. Overlapping leases are rejected.

BANDWIDTH QUOTAS

Each lease enforces a throughput quota per band, preventing congestion without global gas auctions or priority fee markets.

METABOLIC BURN

Continuous value streams burn tokens over elapsed nanoseconds, creating predictable, time-proportional fee economics.

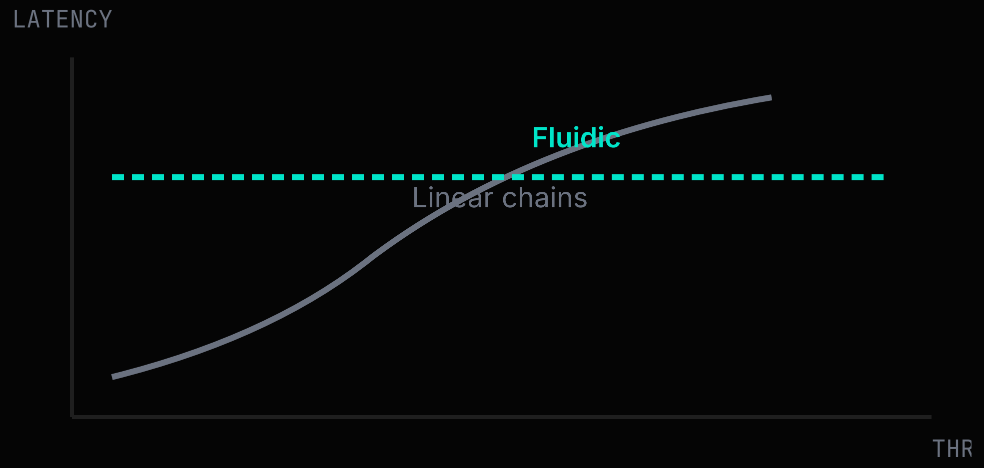
DEX Pool | Agent Swarm | RWA Feed | Unleased spectrum

FREQUENCY BAND ALLOCATION BY STAKED APPLICATION

COMPETITION

Linear Chains Cannot Escape Their Serialization Trade-off

- **Faster finality and bigger blocks** only move the bottleneck; they do not remove it.
- **DAG-based ledgers** improve parallelism but still rely on total ordering for state-dependent spends.
- **Fluidic** is the only architecture that separates commutative aggregation from causal ordering at the mathematics layer.



THE TEAM

Built by Specialists in Cryptography, Signal Processing, and Distributed Systems

CRYPTOGRAPHY

Lattice-based primitives, Ed25519 signature schemes, and BLAKE3 hashing. Security-first design with formal verification targets.

SIGNAL PROCESSING

FFT/NTT algorithm design, prime-field arithmetic, and wave-field synthesis for exact algebraic consensus.

DISTRIBUTED SYSTEMS

BFT consensus research, high-frequency trading infrastructure, and peer-to-peer mesh networking at production scale.

THE ASK

Public Testnet Launch and Cryptography R&D Funding

01 **Public Incentivized Testnet**

Deploy oscillator mesh on public infrastructure with economic incentives for validators and stress-test participants.

02 **Security Audit & Formal Proofs**

Complete liveness and safety proofs for the vector-clock DAG and NTT consensus primitive.

03 **Open Source & Reproducible Benchmarks**

Publish the protocol, test vectors, and reproducible performance benchmarks for community review.

Testnet — Audit — Formal Proofs — Mainnet Candidate

THE FOUNDERS

Founders Combining Cryptographic Rigor with Production Engineering

We have spent years designing consensus primitives, optimizing low-latency distributed systems, and shipping production-grade infrastructure. Fluidic turns that expertise into a new state layer.

TOMAS FEDORKO

Co-founder & Protocol Lead. Architect of the NTT consensus engine, vector-clock DAG, and oscillator mesh. Background in cryptography, prime-field arithmetic, security research.

PETER RAPAVY

Co-founder & Engineering Lead. Built the TCP gossip layer, Docker partition testbed, and continuous benchmarking pipeline. Background in high-frequency trading infrastructure.

CONTACT

Let's build the continuous state layer together.

The benchmarks are reproducible. The mechanism works.

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